

DiT-Distill: Open-Set Fine-Grained Retrieval via Generative Curriculum Knowledge

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Abstract

Open-set fine-grained retrieval (OSFR) is a challenging task where models must generalize to unseen subcategories. Existing methods often fail this, as they embed category-specific semantics from closed-set training labels. Recently, diffusion transformers (DiT) have shown promise by encoding attribute-centric, generative curriculum knowledge that is agnostic to these labels. However, the vanilla DiT is not optimized for fine-grained visual discrepancies and its massive size makes deployment infeasible. To solve this, we propose **DiT-Distill**, a framework to first refine and then distill this knowledge. We introduce a conditional discrepancy refinement strategy to fine-tune the DiT, forcing it to focus on discrepancy-aware, attribute-centric details rather than holistic context. Subsequently, a generative curriculum distillation mechanism transfers this refined, hierarchical knowledge from multiple diffusion timesteps of the DiT into a lightweight backbone using a generative infusion module and a curriculum alignment loss. This process results in an efficient retrieval model that enables DiT-free inference. Extensive experiments show DiT-Distill achieves state-of-the-art performance on open-set fine-grained datasets.

1. Introduction

Open-set fine-grained retrieval (OSFR) aims to retrieve visually similar subcategories that are unseen during training. This task is essential in many real-world applications, such as intelligent transportation (e.g., retrieving different car models [44]) and biodiversity conservation (e.g., identifying endangered species [15]). Given its practical importance, extensive research has focused on learning discriminative and generalizable embeddings that can robustly represent fine-grained visual variations.

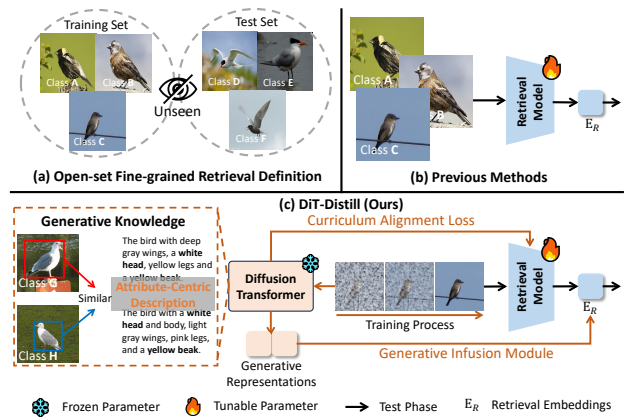


Figure 1. (a) **Definition of OSFR**: retrieving visually similar subcategories unseen during training. (b) **Previous Methods**: Trained in a closed-set, the retrieval model learns representations coupled with predefined category semantics, which limits generalization to unseen subcategories. (c) **Our Solution**: DiT-Distill avoids this limitation by distilling generative curriculum knowledge (agnostic to training labels) from a pre-trained DiT, thereby enabling robust generalization.

Existing OSFR methods can be broadly categorized into two categories: *metric-based* and *localization-based*. Metric-based methods [9, 30, 39] learn embeddings by minimizing intra-class distances while maximizing inter-class distances in a shared feature space. Localization-based methods [17, 42, 43], in contrast, enhance the image encoder to capture subtle visual discrepancies among subcategories, often by focusing on distinctive object parts. However, as illustrated in Fig. 1 (b), both paradigms are trained under a closed-set assumption, where all training categories are predefined. Consequently, their learned representations are tightly coupled with category semantics and tend to generalize poorly to unseen subcategories.

To overcome this limitation, it is desirable to learn representations that capture intrinsic visual attributes, indepen-

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dent of predefined class boundaries. Recent advances in text-to-image diffusion models [24, 33], such as the diffusion transformer (DiT) [24], offer a promising direction [41, 50]. Trained on web-scale image-text pairs, DiT can synthesize visually similar instances across diverse categories guided by textual prompts, without being restricted to specific labels, as shown in Fig. 1 (c). This ability suggests that DiT inherently encodes a rich, attribute-centric knowledge base that may benefit OSFR. Moreover, the denoising process in DiT progresses from coarse to fine: early timesteps capture global structural information, while later ones refine local details. This hierarchical refinement mirrors the human perception process in fine-grained recognition, implying that the diffusion dynamics embed a form of *generative curriculum knowledge* that could enhance discriminative learning.

Despite these advantages, directly employing diffusion models for OSFR remains challenging. Recent studies [5, 14, 47] show that diffusion-based backbones excel in visual perception tasks but underperform discriminative architectures such as ViT [8] in high-level semantic reasoning (e.g., classification [5]). This performance gap arises because the vanilla DiT captures holistic image appearances, including background cues, rather than emphasizing category-relevant discrepancies. Furthermore, the large model size of DiT (often tens to hundreds of times more parameters than discriminative retrieval networks) renders direct deployment impractical. These limitations motivate two key research questions: **(Q1)** How can we guide DiT to focus on fine-grained discrepancies among subcategories instead of holistic image content? **(Q2)** How can we transfer the generative curriculum knowledge of DiT into an efficient discriminative model for DiT-free inference?

To address these questions, we propose **DiT-Distill**, a novel framework that distills *generative curriculum knowledge* (GCK) from a pre-trained text-to-image DiT [24] into a discriminative retrieval network for OSFR, as illustrated in Fig. 2. To solve **Q1**, we design a **Conditional Discrepancy Refinement (CDR)** strategy to enhance DiT’s sensitivity to fine-grained discrepancies. We first leverage multimodal foundation models [1, 3] to construct “context-to-object” image pairs (where the “context” image is the full, original image and the “object” image is a fine-grained, object-centric view) and their corresponding attribute-centric descriptions. A DiT model is then fine-tuned using LoRA [13] via the CDR strategy, where it is conditioned on the full “context” image and the attribute-centric description to generate the “object-centric” view. This strategy forces the DiT to focus its latent space on the visual discrepancies while ignoring the irrelevant surrounding context, effectively “sharpening” its generative knowledge for our task. To solve **Q2**, we introduce a novel **Generative Curriculum Distillation (GCD)** mechanism.

This mechanism efficiently transfers the refined hierarchical knowledge from the CDR-DiT into the lightweight retrieval backbone. Specifically, GCD consists of: (a) a **generative infusion module** that architecturally bridges the generative and discriminative feature spaces, and (b) a **curriculum alignment loss** that progressively distills the GCK by aligning the retrieval model to different diffusion timesteps, starting from coarse-grained knowledge (early steps) and advancing to fine-grained details (later steps). Through this process, the discriminative model “absorbs” DiT’s hierarchical and fine-grained reasoning abilities, allowing for efficient DiT-free inference at test time.

In summary, the main contributions of this work are:

1. We are the first to propose distilling *generative curriculum knowledge* from a text-to-image DiT into a discriminative retrieval framework for OSFR.
2. We propose a *conditional discrepancy refinement* strategy to fine-tune DiT, enabling it to learn context-invariant and discrepancy-aware representations.
3. We develop a *generative curriculum distillation* mechanism that efficiently transfers the refined generative knowledge to a lightweight discriminative model via a novel fusion architecture and alignment paradigm.
4. Extensive experiments demonstrate that DiT-Distill achieves state-of-the-art performance on widely used open-set fine-grained retrieval benchmarks.

2. Related Works

2.1. Open-Set Fine-Grained Retrieval

Existing methods for open-set fine-grained retrieval (OSFR) can be roughly divided into two categories: metric-based and localization-based. Metric-based methods [9, 16, 21, 27, 30, 39] aim to learn an embedding space where samples from the same subcategory are pulled closer together and those from different subcategories are pushed apart. Although these studies have achieved impressive results, they mainly optimize image-level features, which often contain noisy and non-discriminative details. To address this, localization-based methods [4, 17, 18, 20, 36, 43] train subnetworks to locate discriminative regions or design strategies to extract informative object features for retrieval. In particular, FRPT [43] and DVA [18] adapt frozen pre-trained models to capture category-specific discrepancies for OSFR through sample-level and feature-level adaptation. However, these methods rely on predefined subcategories to represent similar objects, which embed category-specific semantics into the retrieval representations and limit generalization to unseen subcategories. To overcome this limitation, we propose DiT-Distill, which distills attribute-centric, generative curriculum knowledge in a pre-trained diffusion transformer during its diffusion process to a retrieval model trained in closed-set scenarios.

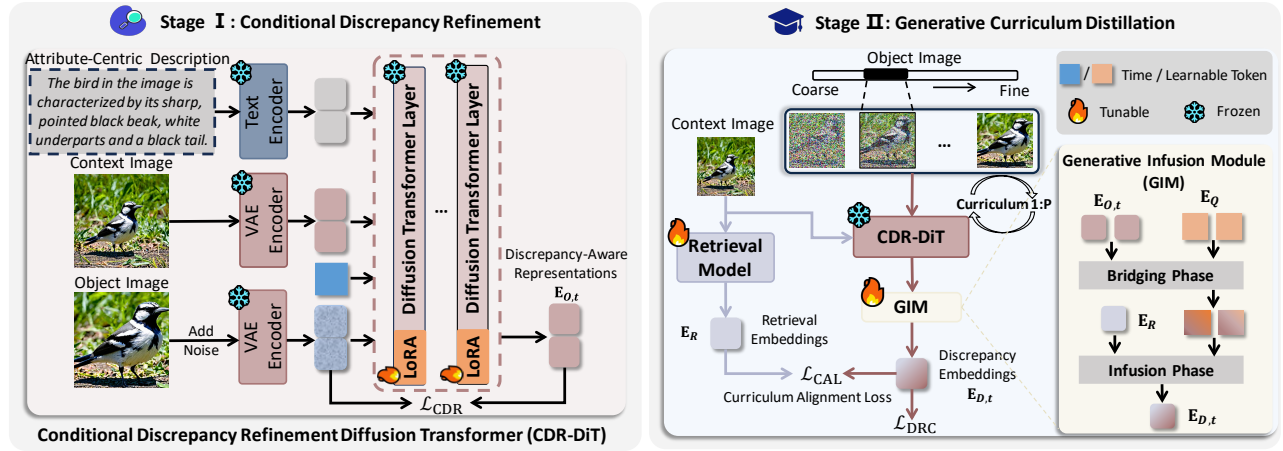


Figure 2. Overview of DiT-Distill. The framework consists of two stages. In Stage I, a Conditional Discrepancy Refinement Diffusion Transformer (CDR-DiT) is trained to make DiT more sensitive to fine-grained visual discrepancies. In Stage II, the refined generative knowledge from CDR-DiT is transferred into a compact retrieval model using a generative curriculum distillation mechanism.

2.2. Diffusion Model for Discriminative Tasks

Recently, diffusion models [24–26, 33] have attracted significant attention for their generative capabilities. Numerous studies [10, 50] have explored extending their applications to a wide range of discriminative tasks, such as detection, segmentation, and classification, *etc.* For diffusion-based discriminative tasks, one category of methods [6, 19, 31] reformulates the tasks as progressive denoising from random noise, as seen in DiffusionDet [6] and DDP [14]. Another method uses a denoising autoencoder pre-trained on text-to-image generation as the backbone for discriminative tasks [11, 35, 37, 38, 48]. For example, DIFT [38] and VPD [48] employ the pre-trained UNet in stable diffusion [33] as a feature extractor for visual correspondence and dense prediction tasks, respectively. Although these methods have achieved promising results in visual perception tasks, they remain less effective than discriminative models in highly semantic tasks such as classification [5]. Moreover, the vanilla diffusion model is not optimized for fine-grained visual discrepancies, and its large size makes deployment impractical. To address this, we design a conditional discrepancy refinement strategy that enables a pre-trained diffusion transformer to capture subtle discrepancies in fine-grained objects and then transfer discrepancies to an efficient discriminative model.

3. Methodology

In this section, we present **DiT-Distill**, a novel framework to distill *generative curriculum knowledge* (GCK) from a pre-trained diffusion transformer (DiT) into an efficient discriminative retrieval model for OSFR. The overall architecture is illustrated in Fig. 2.

Our methodology is structured as follows. We first introduce the fundamental concepts of the diffusion transformer

and the discriminative retrieval network in Sec. 3.1. The core of our method is then designed to solve the two key challenges identified in the Introduction. First, Sec. 3.2 details the proposed *Conditional Discrepancy Refinement* (CDR) strategy to solve (Q1). This strategy fine-tunes the pre-trained DiT to refine its GCK, forcing it to focus on subtle, object-centric details. Second, Sec. 3.3 describes the *Generative Curriculum Distillation* (GCD) mechanism to solve (Q2). This mechanism efficiently transfers the refined generative knowledge from the CDR-DiT into a compact retrieval model. As we will detail, the GCD mechanism consists of two complementary components: (1) a *generative infusion module* that architecturally enriches retrieval embeddings, and (2) a *curriculum alignment loss* that distills hierarchical knowledge for efficient *DiT-free inference*. Finally, Sec. 3.4 summarizes the overall training pipeline.

3.1. Network Architecture: Student and Teacher

Our DiT-Distill framework is built on a “Student-Teacher” architecture. We now define the two core components: the lightweight backbone that serves as the “student” and the DiT that serves as the “teacher”.

Discriminative Backbone (The Student). The retrieval model is our lightweight discriminative backbone and the ultimate *target of distillation*. Given an input image $\mathbf{I} \in \mathbb{R}^{3 \times H \times W}$, the model encodes it into a set of patch embeddings $\mathbf{E} = [\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_N] \in \mathbb{R}^{N \times C}$ through a learnable linear projection, where C is the embedding dimension. A class token $\mathbf{E}_{\text{cls}} \in \mathbb{R}^C$ is concatenated and jointly processed by L transformer layers. After L layers, the updated class token $\mathbf{E}_{\text{cls},L}$ is taken as the final retrieval embedding, denoted by $\mathbf{E}_R$. This compact, discriminative embedding $\mathbf{E}_R$ is what we aim to enrich with generative knowledge, enabling efficient, **DiT-free inference** at test time.

Generative Knowledge Source (The Teacher). The diffu-

sion transformer (DiT) [24] serves as the *source of the GCK* we aim to distill. This knowledge is not arbitrary; it is inherently embedded in the DiT’s hierarchical denoising process, which naturally progresses from coarse-grained global structures (in early timesteps) to fine-grained, attribute-centric details (in later timesteps). Technically, this process is formulated as follows. The DiT operates in a latent space, where a VAE $\mathcal{E}$ first encodes the image $\mathbf{I}$ into a latent $\mathbf{X} = \mathcal{E}(\mathbf{I})$. During forward diffusion, this latent is interpolated with noise ϵ to create a noisy state $\mathbf{X}_t$ at timestep $t \in [0, 1]$ as $\mathbf{X}_t = t\mathbf{X} + (1 - t)\epsilon$. The DiT, parameterized by v_θ , is then trained on a denoising objective to predict the clean data’s trajectory:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, \mathbf{X}, \epsilon} \left[\left| (\mathbf{X} - \epsilon) - v_\theta(\mathbf{X}_t, t) \right|^2 \right]. \quad (1)$$

Crucially, this objective forces v_θ to learn a structured, hierarchical representation of the visual world. It is this internal, coarse-to-fine knowledge (the GCK) that we will first refine (Sec. 3.2) and then distill (Sec. 3.3).

3.2. Conditional Discrepancy Refinement

Visual discrepancies play a critical role in fine-grained understanding under open-set conditions [45, 46]. However, as identified in (Q1), the vanilla diffusion transformer (DiT) primarily models holistic image appearances rather than emphasizing the subtle local differences that distinguish visually similar objects. To address this, we propose a **Conditional Discrepancy Refinement (CDR)** strategy. This strategy fine-tunes the pre-trained DiT to focus on fine-grained visual discrepancies, thereby refining its generative curriculum knowledge for the OSFR task.

To create the training data for this strategy, we first construct *context-to-object* image pairs and corresponding attribute-centric textual descriptions. Given a context (original) image $\mathbf{I}$, the open-vocabulary detector GroundingDINO [29] identifies the fine-grained object using its super-category name (e.g., “bird”) and extracts an object-centric image $\mathbf{I}_O$ that highlights the object while suppressing irrelevant background regions. Low-quality crops are discarded. Next, this object-centric image $\mathbf{I}_O$ is fed into the multimodal large language model Qwen2.5-VL-7B [1] with an attribute-focused instruction: “Describe the [cls] in the image and its characteristics in one sentence, without outputting its category name.”. This process generates an attribute-centric textual description $\mathbf{T}_{\text{text}}$ that provides explicit, non-category-specific guidance for the model’s refinement.¹

We then build a *Conditional Discrepancy Refinement DiT* (CDR-DiT) by fine-tuning the pre-trained DiT with LoRA [13]. The model is tasked to generate the object ($\mathbf{I}_O$)

while being conditioned on both the context ($\mathbf{I}$ and the text $\mathbf{T}_{\text{text}}$). At diffusion timestep t , the CDR-DiT v_θ receives all three inputs:

$$\begin{aligned} \mathbf{E}_I, \mathbf{E}_{O,t} &= v_\theta(\mathbf{T}_{\text{text}}, \mathbf{X}_I, \mathbf{X}_{O,t}, t), \\ \mathbf{X}_{O,t} &= t\mathcal{E}(\mathbf{I}_O) + (1 - t)\epsilon, \\ \mathbf{X}_I &= \mathcal{E}(\mathbf{I}), \end{aligned} \quad (2)$$

where $\mathcal{E}$ is the VAE encoder, $\epsilon \sim \mathcal{N}(0, 1)$, $\mathbf{X}_{O,t}$ is the noisy object latent, and $\mathbf{X}_I$ is the clean context latent. The CDR-DiT is optimized via a direct reconstruction objective on the object latent:

$$\mathcal{L}_{\text{CDR}} = \mathbb{E}_{t, \mathbf{X}_O, \epsilon} \left[\left| (\mathbf{X}_O - \epsilon) - \mathbf{E}_{O,t} \right|^2 \right]. \quad (3)$$

This task is deceptively challenging and is the key to our solution. To successfully reconstruct $\mathbf{X}_O$ (the object) from its noisy version $\mathbf{X}_{O,t}$, while also being conditioned on $\mathbf{X}_I$ (the full context), the model v_θ is forced to learn to separate object-centric information from background-centric information. It must implicitly learn to “subtract” the context features present in $\mathbf{X}_I$ and focus only on the attribute-centric features (guided by $\mathbf{T}_{\text{text}}$) that define the object. This process compels DiT to refine its internal GCK, making it highly sensitive to the object-specific discrepancies.

After this refinement, the CDR-DiT’s internal representations $\mathbf{E}_{O,t}$ are no longer just holistic features; they are truly discrepancy-aware, as they are the product of a model explicitly trained to distinguish object from context. These refined, attribute-centric features serve as the high-quality “teacher” knowledge to be transferred in the subsequent distillation stage.

3.3. Generative Curriculum Distillation

As identified in (Q2), a key challenge of using the DiT lies in its massive scale (up to 12 billion parameters), which makes direct deployment computationally infeasible. To solve this, we propose the **Generative Curriculum Distillation (GCD)** mechanism, which efficiently transfers the refined knowledge from our CDR-DiT (Sec. 3.2) into the lightweight retrieval backbone (Sec. 3.1).

This mechanism has two key parts: (1) the **Generative Infusion Module (GIM)**, an architectural bridge that connects the teacher (DiT) and student (ViT) feature spaces, and (2) the **Distillation Objectives**, a set of loss functions that govern the knowledge transfer. Together, they enable the retrieval model to inherit the DiT’s fine-grained reasoning abilities, allowing for efficient **DiT-free inference**.

3.3.1. Generative Infusion Module

To enable distillation, we must first create a pathway for information to flow from the generative “teacher” to the discriminative “student”. The GIM serves as this architectural bridge. As shown in Fig. 2 (upper right), the GIM is a lightweight module that operates in two sequential phases:

¹Additional details on prompt design and filtering are provided in the supplementary material.

Bridging phase. First, to create a trainable “adapter” for the generative knowledge, we introduce k learnable query embeddings $\mathbf{E}_Q \in \mathbb{R}^{k \times C}$. These queries “interrogate” the frozen, discrepancy-aware representations $\mathbf{E}_{O,t}$ (from the CDR-DiT) using a *bridging encoder* $\mathcal{E}_b$ composed of n cross-attention layers as $\hat{\mathbf{E}}_{Q,t} = \mathcal{E}_b(\mathbf{E}_Q, \mathbf{E}_{O,t})$. The output $\hat{\mathbf{E}}_{Q,t}$ now holds the essential generative, attribute-centric cues from the teacher for that specific timestep t .

Infusion phase. Second, these adapted generative cues $\hat{\mathbf{E}}_{Q,t}$ are “infused” into the student’s retrieval embedding $\mathbf{E}_R$ (from the backbone). Both are fed into an *infusion encoder* $\mathcal{E}_f$, which uses self-attention to fuse them into a single, enhanced representation as $\mathbf{E}_{D,t}, \hat{\mathbf{E}}'_{Q,t} = \mathcal{E}_f(\mathbf{E}_R, \hat{\mathbf{E}}_{Q,t})$. The resulting $\mathbf{E}_{D,t}$ is our final *discrepancy-enhanced embedding* for timestep t . This embedding is now “aware” of both the student’s discriminative knowledge and the teacher’s generative knowledge. It is this enhanced embedding that we will use to compute our training objectives.

3.3.2. Distillation Objectives

To define the curriculum, we first select a total number of diffusion steps T for our sampling schedule (where $p = 1$ is the noisiest and $p = T$ is the clearest). We found $T = 28$ to be an effective and efficient choice for the FLUX [24] scheduler, balancing generation quality with speed. Given the set of chosen curriculum timesteps $\mathcal{P} = \{10, 16, 22, 28\}$, we train the retrieval backbone and the GIM module using a combined objective $\mathcal{L}$:

$$\mathcal{L} = \mathcal{L}_{\text{TASK}} + \alpha \mathcal{L}_{\text{ALIGN}}, \quad (4)$$

where α is a weighting coefficient. This objective forces the retrieval model to learn two things simultaneously: **what** to retrieve (the task) and **how** to think (the knowledge).

Task-Learning Objective ($\mathcal{L}_{\text{TASK}}$). First, the enhanced embedding $\mathbf{E}_{D,t_p}$ at each curriculum stage p must be optimized for the retrieval task. To prevent overfitting to the closed-set training labels, we first define the proxy-based retrieval loss as a *Discrepancy Representation Constraint (DRC)* to force embeddings to cluster around learnable category proxies $\mathbf{P}_m \in \mathbb{R}^C$ for a single timestep t_p :

$$\mathcal{L}_{\text{DRC}} = -\log \left(\frac{\exp(d(\mathbf{E}_{D,t_p}, \mathbf{P}_m)/\tau)}{\sum_{m' \in \mathcal{M}} \exp(d(\mathbf{E}_{D,t_p}, \mathbf{P}_{m'})/\tau)} \right), \quad (5)$$

where $d(\cdot, \cdot)$ denotes cosine similarity, τ is a temperature parameter, $\mathbf{P}_m$ is the proxy for the sample’s correct class m , and $\mathcal{M}$ is the set of all C_{train} proxies in the training set. The full $\mathcal{L}_{\text{TASK}}$ is then the expectation of this loss over all P stages in our curriculum $\mathcal{P}$:

$$\mathcal{L}_{\text{TASK}} = \mathbb{E}_{p \in \mathcal{P}} [\mathcal{L}_{\text{DRC}}(\mathbf{E}_{D,t_p}, \mathbf{P}_m)]. \quad (6)$$

Knowledge-Alignment Objective ($\mathcal{L}_{\text{ALIGN}}$). Second, we must ensure the student’s own internal representation $\mathbf{E}_R$

(the one used at test time) “absorbs” all the rich knowledge contained in the enhanced embedding $\mathbf{E}_{D,t_p}$. We achieve this using our *Curriculum Alignment Loss (CAL)*, which acts as a strong consistency regularizer, also computed over the curriculum:

$$\mathcal{L}_{\text{ALIGN}} = \mathcal{L}_{\text{CAL}} = \mathbb{E}_{p \in \mathcal{P}} [\|\mathbf{E}_R - \mathbf{E}_{D,t_p}\|_F^2], \quad (7)$$

where $\|\cdot\|_F$ is the Frobenius norm. This loss creates a symmetric “pull” between the student’s output ($\mathbf{E}_R$) and the fused output ($\mathbf{E}_{D,t_p}$). The gradients from $\mathcal{L}_{\text{ALIGN}}$ flow back to all trainable parameters including the student backbone (which produces $\mathbf{E}_R$ and also contributes to $\mathbf{E}_{D,t_p}$) and the GIM module (which also contribute to $\mathbf{E}_{D,t_p}$). This forces the student and the GIM to mutually align.

By training with $\mathcal{L}_{\text{TASK}}$ and $\mathcal{L}_{\text{ALIGN}}$ simultaneously, the student backbone ($\mathbf{E}_R$) is forced to “internalize” the generative knowledge, which is the only way to satisfy both losses and ultimately enables efficient DiT-free inference.

3.4. Training Pipeline and DiT-Free Inference

We train the DiT-Distill framework in two sequential stages, which are designed to directly correspond to the two challenges outlined in our methodology.

Stage I: Solving (Q1) via CDR-DiT Refinement. In the first stage, we **solve (Q1)** by creating our expert “teacher” model. We execute the *Conditional Discrepancy Refinement (CDR)* strategy (Sec. 3.2) by fine-tuning the pre-trained DiT using LoRA [13] with the $\mathcal{L}_{\text{CDR}}$ objective. This step yields the **CDR-DiT**, a specialized generative model whose GCK is now “sharpened” to be highly sensitive to fine-grained, attribute-centric visual discrepancies.

Stage II: Solving (Q2) via GCD and DiT-Free Inference. In the second stage, we **solve (Q2)** by distilling the “teacher’s” knowledge into the “student”. The CDR-DiT’s parameters are frozen, acting as a fixed knowledge source. We then train the lightweight retrieval backbone and the generative infusion module by performing *Generative Curriculum Distillation* (Sec. 3.3). The model is trained with the combined distillation objectives defined in Eq. (4). This two-part objective, computed across the full curriculum $\mathcal{P}$, forces the lightweight backbone to simultaneously learn the retrieval task ($\mathcal{L}_{\text{TASK}}$) while “internalizing” the teacher’s reasoning patterns ($\mathcal{L}_{\text{ALIGN}}$). This distillation is the key to our final goal. By forcing the student to “absorb” the teacher’s knowledge, the teacher model becomes unnecessary after training. Consequently, the entire DiT branch is discarded. During evaluation, only the lightweight retrieval model $\mathcal{E}_R$ is used to generate embeddings, *i.e.*, $\mathbf{E}_R = \mathcal{E}_R(\mathbf{I})$. This achieves our goal of efficient and scalable **DiT-free inference** while preserving the powerful fine-grained reasoning capabilities of the generative model.

Table 1. Comparison with state-of-the-art methods on four fine-grained datasets: CUB-200-2011, Stanford Cars, Stanford Dogs, and NABirds. The best result is shown in **bold**, and the second-best result is underlined.

Methods	CUB-200-2011				Stanford Cars				Stanford Dogs				NABirds			
	R@1	R@2	R@4	R@8	R@1	R@2	R@4	R@8	R@1	R@2	R@4	R@8	R@1	R@2	R@4	R@8
<i>Baseline “Student” Backbone (Ours)</i>																
ViT-B/16 (Baseline)	77.4	85.4	90.7	94.5	72.8	82.3	89.0	93.3	82.9	89.8	94.1	96.5	72.0	80.6	86.8	91.5
<i>SOTA Discriminative Methods (Prior Art)</i>																
PNCA [30]	49.2	61.9	67.9	72.4	73.2	82.4	86.4	88.7	77.3	85.6	91.4	95.3	45.2	56.5	66.7	76.0
DGCR [49]	67.9	79.1	86.2	91.8	75.9	83.9	89.7	94.0	-	-	-	-	-	-	-	-
DAS [28]	69.2	79.3	87.1	92.6	87.8	93.2	96.0	97.9	-	-	-	-	-	-	-	-
PNCA++ [39]	70.1	80.8	88.7	93.6	90.1	94.5	97.0	98.4	86.0	92.3	95.6	<u>97.7</u>	63.4	74.0	82.2	88.4
IBC [34]	70.3	80.3	87.6	92.7	88.1	93.3	96.2	98.2	-	-	-	-	-	-	-	-
Proxy-Anchor [22]	71.1	80.4	87.4	92.5	88.3	93.1	95.7	97.0	86.1	91.9	95.7	97.6	54.3	64.9	74.5	82.2
HIST [27]	71.4	81.1	88.1	-	89.6	93.9	96.4	-	86.2	92.3	95.6	97.5	51.8	62.8	72.5	81.0
FRPT [43]	74.3	83.7	89.8	94.3	<u>91.1</u>	<u>95.1</u>	<u>97.3</u>	<u>98.6</u>	-	-	-	-	-	-	-	-
DFML [40]	79.1	86.8	-	-	89.5	93.9	-	-	-	-	-	-	-	-	-	-
Hyp-ViT [9]	84.0	90.2	94.2	96.4	86.0	91.9	95.2	97.2	<u>87.8</u>	<u>92.8</u>	<u>95.9</u>	97.6	<u>79.2</u>	<u>86.7</u>	<u>92.1</u>	<u>95.3</u>
DVA [18]	<u>84.9</u>	<u>90.6</u>	<u>94.5</u>	<u>96.7</u>	90.7	94.8	97.1	98.4	-	-	-	-	-	-	-	-
DiT-Distill (Ours)	87.2	92.4	95.2	97.0	91.4	95.2	97.5	98.9	89.4	93.7	96.4	97.8	83.7	89.8	93.9	96.2

4. Experiments

4.1. Experimental Setup

Datasets. **CUB-200-2011** [2] contains 11,788 images of 200 bird species. Following standard protocol [43], we use the first 100 species (5,864 images) for training and the remaining 100 species (5,924 images) for testing. **Stanford Cars** [23] includes 16,185 images from 196 car variants. The first 98 variants (8,054 images) are used for training, while the remaining 98 variants (8,131 images) form the test set. **Stanford Dogs** [7] comprises 20,580 images covering 120 dog subcategories. We use the first 60 subcategories (10,651 images) for training and the remaining 60 subcategories (9,929 images) for testing. **NABirds** [12] is a larger dataset with 48,562 images of 555 North American bird species. We adopt the split from prior work [17], using the first 255 categories (20,984 images) for training and the remaining 300 categories (27,578 images) for testing.

Evaluation Protocols. We evaluate retrieval performance using Recall@K with cosine distance, following previous work [18]. For each query image in the test set, the top-K most similar images are retrieved based on feature similarity. A query achieves a recall score of 1 if at least one positive image (i.e., from the same subcategory) appears within the top-K results; otherwise, the score is 0. The overall Recall@K is obtained by averaging over all queries.

Implementation Details. We adopt ViT-B/16 [8] pre-trained on ImageNet-21K [32], as the retrieval backbone, and FLUX [24] as the pre-trained diffusion transformer. All input images are resized to 256×256 and center-cropped to 224×224 . In the *Stage I*, We fine-tune the DiT using LoRA [13] with a rank of 16. The Adam optimizer is used with a constant learning rate of 1×10^{-4} . Training is conducted for 30,000 steps with a batch size of 1. In the *Stage II*, we freeze the CDR-DiT and train the remain-

Table 2. Ablation study on the components of DiT-Distill on CUB-200-2011. We analyze the step-by-step impact of our framework on accuracy (R@1) and inference speed (Time).

Model Configuration	$\mathcal{L}_{CDR}$	GIM	$\mathcal{L}_{CAL}$	R@1	Time
① ViT “Student”				77.4%	2.6ms
② Teacher-Assisted (Vanilla)		✓		85.3%	33.5ms
③ Teacher-Assisted (Refined)	✓	✓		86.5%	33.5ms
④ Teacher-Free (Ours)	✓	✓	✓	87.2%	2.6ms

ing modules of DiT-Distill using the Adam optimizer with a cosine-annealing learning rate schedule. The initial learning rate is 1×10^{-3} for all datasets. Our distillation curriculum, $\mathcal{P}$, is defined by $P = 4$ stages, sampled at timesteps $p \in \{10, 16, 22, 28\}$. Training is performed for 30 epochs on CUB-200-2011, Stanford Cars, and Stanford Dogs, and for 15 epochs on NABirds, all with a batch size of 32. All experiments are conducted on a single NVIDIA A800 GPU.

4.2. Comparisons with the State-of-the-Arts

As shown in Table 1, our DiT-Distill achieves state-of-the-art results, significantly outperforming all prior methods. This victory is driven by our core distillation strategy. The most critical comparison is against our “Student” baseline (ViT-B/16), which quantifies the impact of our framework. Our method boosts the R@1 performance from 77.4% to 87.2% (+9.8%) on CUB, and even more dramatically from 72.8% to 91.4% (+18.6%) on Stanford Cars. This massive gain on the exact same backbone provides direct proof that our generative curriculum distillation mechanism (solving Q2) successfully “teaches” the lightweight student model. This distillation-based gain allows us to surpass specialized SOTA methods. Prior methods like Hyp-ViT [9] and DVA [18] are trained to model category-specific discrepancies. Our model, in contrast, learns from the DiT’s GCK, which is label-agnostic and refined by our CDR (Q1) stage.

Table 3. Ablation studies on the CUB-200-2011 dataset. (a) Different numbers of learnable embeddings in $\mathcal{E}_b$. (b) Different numbers of layers in the bridging encoder. (c) Different training stages of curriculum alignment loss. The best results are shown in **bold**.

(a) Number of Learnable Embeddings (k)					(b) Layers in Bridging Encoder (n)					(c) Timestep Curriculum (t_p)				
Number (k)	4	16	32	64	Layer (n)	0	1	3	6	Curriculum	t_4	t_3-t_4	t_2-t_4	t_1-t_4
Recall@1	86.5%	86.9%	87.2%	87.0%	Recall@1	85.1%	86.3%	87.2%	86.7%	Recall@1	86.7%	87.0%	87.2%	87.2%
Recall@2	91.8%	92.2%	92.4%	92.3%	Recall@2	90.9%	91.7%	92.4%	92.1%	Recall@2	92.0%	92.3%	92.4%	92.4%

Table 4. Robustness to data source on CUB-200-2011. We compare our **DiT-Distill** framework when trained using two different data sources for its CDR stage.

CDR Data Source	R@1	R@2
Oracle (Manual Bounding Box)	87.4%	92.5%
Detector-based (Ours)	87.2%	92.4%

This attribute-centric GCK generalizes far more effectively to unseen subcategories, explaining our large performance leads on the highly diverse Cars and NABirds datasets.

4.3. Ablation Studies

Effect of Distillation Components. We analyze the components of DiT-Distill in Table 2. (1) Baseline (①): The ViT “Student” baseline is fast but its performance is limited. (2) Teacher-Assisted (②): By adding the GIM (trained with $\mathcal{L}_{\text{TASK}}$) and using the vanilla DiT at inference, performance leaps to 85.3% R@1. This proves the generative knowledge is highly valuable, but also that this “teacher-assisted” model is impractically slow (33.5ms). (3) Refined Teacher-Assisted (③): By swapping the vanilla teacher for our CDR-refined teacher, performance is further boosted to 86.5% R@1. This directly validates our (Q1) solution, proving that the CDR strategy creates a measurably “smarter” teacher. The model remains impractically slow. (4) DiT-Distill (④): Finally, adding the crucial $\mathcal{L}_{\text{CAL}}$ (distillation loss) solves (Q2). It allows us to discard the DiT, returning inference time to an efficient 2.6ms while also further increasing performance to a state-of-the-art 87.2% R@1. This validates our full distillation framework, proving it is both highly accurate and efficient.

Analysis of GIM Hyperparameters. We ablate the key hyperparameters for our Generative Infusion Module (GIM) in Table 3 (a) and (b). First, for the number of learnable queries (k), we find in Table 3 (a) that performance peaks at $k = 32$. A smaller number ($k = 16$) appears to lack the capacity to fully capture the generative knowledge, while a larger number ($k = 64$) shows a slight performance drop, likely due to overfitting on the closed-set proxy task. We therefore set $k = 32$ as the optimal trade-off. Next, we evaluate the depth (n) of the bridging encoder in Table 3 (b). Using a simple MLP ($n = 0$) is clearly suboptimal (85.1% R@1), as it fails to effectively bridge the generative-discriminative feature gap. Performance increases significantly up to $n = 3$ layers (87.2% R@1). However, at

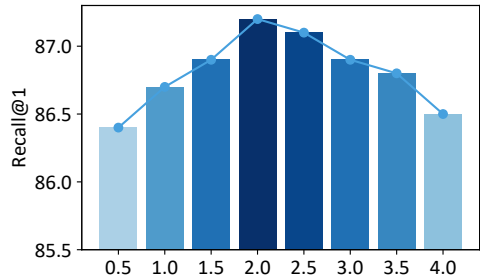


Figure 3. Analysis of the distillation weight α , which balances $\mathcal{L}_{\text{TASK}}$ and $\mathcal{L}_{\text{ALIGN}}$. Results are R@1 on CUB-200-2011.

$n = 6$, performance degrades (86.7%), suggesting that the deeper module begins to overfit or add unnecessary complexity. We thus set $n = 3$ for all experiments.

Analysis of Curriculum Alignment Stages. Finally, we analyze the composition of our curriculum distillation in Table 3 (c). We define four potential stages: $t_4 = 28$ (clearest), $t_3=22$, $t_2=16$, and $t_1=10$ (noisiest). Distilling from only the final, clearest timestep (t_4) yields the worst result (86.7% R@1). Performance steadily improves as we “add back” the coarser, noisier timesteps: t_3-t_4 (87.0%) and t_2-t_4 (87.2%). This result validates our core GCK hypothesis that the coarse, early-timestep knowledge is crucial for learning robust representations. However, adding the noisiest timestep (t_1) provides no additional benefit (87.2%), as t_1 (see Fig. ??) is likely pure noise. We therefore use the curriculum t_2-t_4 as our default.

Robustness to Bounding Box Noise. We now justify our choice of using an open-vocabulary detector for the CDR (Sec. 3.2) data pipeline. A key question is whether our framework requires perfect, human-labeled data to work. To test this, we compare two models in Table 4: (1) an “Oracle” model trained using ground-truth manual bounding box annotations, and (2) our final model, trained using the (imperfect) output of an automatic detector. The Oracle model, with its perfect data, achieves an upper-bound performance of 87.4% R@1. In remarkable contrast, our detector-based model achieves 87.2% R@1. This virtually identical result is a key finding: it demonstrates that our CDR $\rightarrow$ GCD pipeline is highly robust to the object localization noise introduced by the detector. This robustness is a critical feature, as it validates that our framework can be deployed “in the wild” without any dependency on the expensive, manual bounding box annotations.

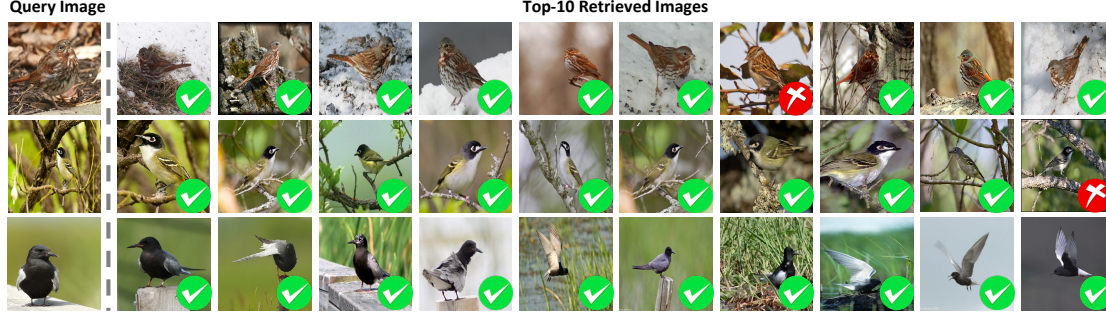


Figure 4. Examples of top-10 retrieved images on CUB-200-2011 by our DiT-Distill.

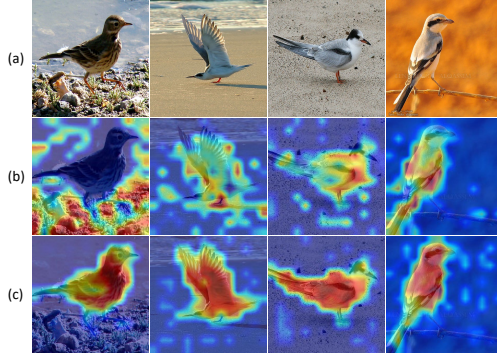


Figure 5. Illustration of discriminative activation maps. (a) is the input image, (b)(c) denote discriminative activation maps generated by the baseline model and DiT-Distill, respectively.

Analysis of Distillation Weight (α). We analyze the sensitivity of our key hyperparameter α in Fig. 3, which controls the trade-off between our two main objectives (Eq. 4). When α is too low, the model prioritizes the retrieval task ($\mathcal{L}_{\text{TASK}}$) but fails to internalize the teacher’s GCK, leading to suboptimal generalization. When α is too high, the model focuses too heavily on mimicking the teacher ($\mathcal{L}_{\text{ALIGN}}$) and “forgets” the primary objective, again hurting performance. The performance peak at $\alpha = 2.0$ validates our approach, as it represents the optimal balance where the “student” model learns both the retrieval task and the teacher’s reasoning.

4.4. Visualization

Validating CDR: Where to Look. In Fig. 5, we first validate that our distillation process solves (Q1) by comparing the discriminative activation maps from a baseline ViT (trained without GCK) to our final DiT-Distill model. The baseline model, lacking refined knowledge, often gets confused by background clutter or non-essential parts of the object (e.g., columns 2 and 3). In stark contrast, our DiT-Distill model, having absorbed the “teacher’s” focus, consistently and tightly localizes the most discriminative, attribute-centric parts of the object, even for unseen subcategories. This visually confirms that our framework successfully “teaches the student where to look”.

Validating GCD: A Better Embedding Space. Second,

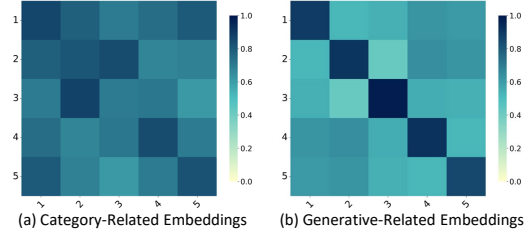


Figure 6. The nearest description for category-related embeddings by baseline model and generative-related embeddings by DiT-Distill, with their similarity shown in grids.

we validate the effectiveness of our GCD mechanism (Q2). In Fig. 6, we compare the embedding space structure by plotting the similarity matrices for five novel subcategories from the baseline (left) and our DiT-Distill (right). The baseline’s “category-related” embeddings show significant off-diagonal noise, indicating poor inter-class separation. Our “generative-distilled” embeddings exhibit far stronger intra-class compactness (the dark diagonal blocks) and superior inter-class separation. This superior embedding structure, learned via GCD, directly translates to the robust retrieval performance shown in Fig. 4. Our model successfully identifies visually similar birds despite large variations in pose and background, proving the student has effectively inherited the teacher’s fine-grained reasoning ability.

5. Conclusion

In this paper, we present DiT-Distill, a framework to solve open-set fine-grained retrieval by distilling generative curriculum knowledge from a DiT. Our method first **refines** the DiT’s knowledge to be discrepancy-aware via a conditional discrepancy refinement strategy. Then, our generative curriculum distillation mechanism transfers this refined, hierarchical knowledge into a lightweight, DiT-free backbone. Our experiments validate a key thesis: generative models can successfully “teach” attribute-centric, generalizable reasoning to compact discriminative models. This generative distillation paradigm offers a promising new direction for fine-grained understanding.

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